An actual 160-bit point-order certificate

Lean proves primality and 160-bit bounds for the explicit modulus p below, nonsingularity of the corrected curve, exact prime order p of an actual Mathlib point G, and the fixed known-answer relation Q=dG. The proof uses 35 Lucas nodes and 448 candidate point nodes. Supplied coordinates and slopes are independently checked by the kernel and connected to actual point addition. [pcc:unverified]

$$E:\ y^2 = x^3 + ax + b, \qquad 2^{159} \le p < 2^{160}, \qquad \mathrm{ord}(G) = p, \quad Q = [d]G.$$

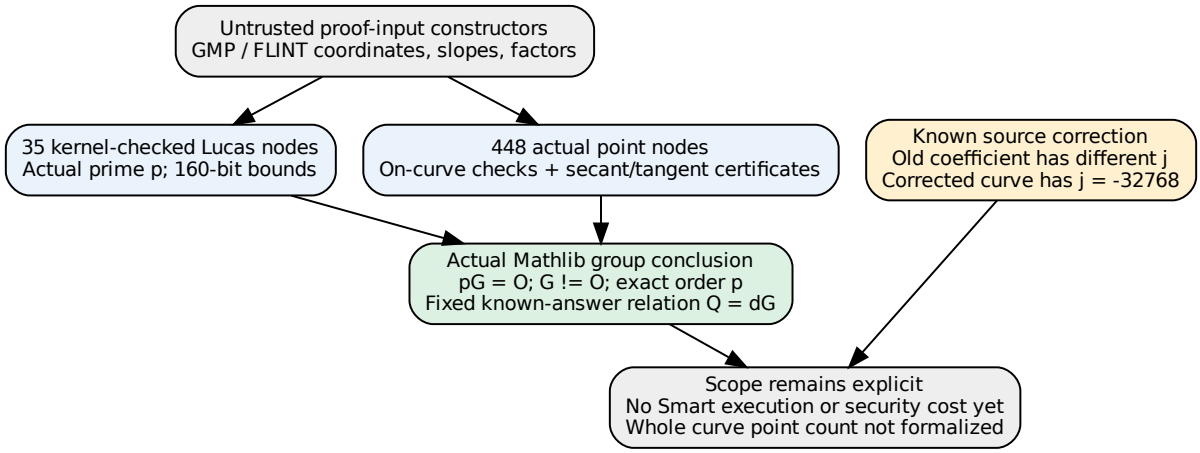


Figure 1: Mathematical witnesses, actual point proofs and explicit scope. [pcc:unverified]

The literal curve from the uncorrected author preprint has a different j-invariant from the stated construction. The author's published erratum already corrects the missing leading digit and replaces its points. Lean reproduces that known discrepancy and checks j=-32768 on the corrected curve; no new error or refutation of the corrected paper is claimed. [pcc:unverified]

Fixed parameters, proof map and next obligation

```
p = 730750818665451459112596905638433048232067471723
a = 42570641384221105410270023816413353830216917647 4
b = 20336293654882693667326444498286633995326553016 6
G = (1, 31053646893989969371896235433899665538136756902 0)
Q = (25947971315827375705036719053485807259422322762 7,
     39414799304662693276020416800594933604368475811)
d = 21039328796666075659613264317217264040508553617 9
```

G and Q are fresh known-answer points on the corrected curve, not the example points printed in either paper version. The actual equality pG=O and G≠O give exact order p because p is prime. This component does not prove the whole curve point count is p, execute Smart's attack, prove its p-adic bridge, or measure security cost. A subsequent concrete attack certificate must be bound to this actual point relation. [pcc:unverified]

Module	Checked obligation
AnomalousPrime	Recursive Lucas primality certificates and bit bounds
CurveData	Actual discriminants, j-invariants and known preprint control
AffineCertificates	Actual secant/tangent group addition from checked slopes
PointTrace	Every numerical point/edge and the scalar chains for pG and dG

Main binding: `AnomalousPointTraceAudit.large_order_audit_certificate`. Full gate: 59 public theorem declarations, zero failing and zero trusted-native exceptions; private generated certificates are compiled and checked transitively. Lean 4.35.0 / Mathlib 56a00585; only propext, Classical.choice, Quot.sound. No holes, custom axioms, native decision exceptions, linter waivers or added elaboration option. Failed sources/logs are retained losslessly; fresh lean4checker is unavailable. [pcc:unverified]

Source commit `c10e04d5` precedes packet `PCC-20261110-90623b`. Actual local verification and signed standing are recorded separately. Proof map and provenance: `../../README.md`, `../../point-sources.json`, `../../point-verification.json`. C++/GMP/FLINT produces untrusted proof inputs; the source writer performs no curve arithmetic. No timing or population inference, native attack run or paid compute in this component. [pcc:unverified]

Sources read: original Smart report HPL-97-128 (1997); original author preprint and two-page erratum for Leprévost et al., IPL 93(5), 225–230 (2005), DOI 10.1016/j.ipl.2004.11.008. Original example and both erratum pages were visually inspected. NIST SP 800-186 explicitly excludes anomalous curves and requires large embedding degree; its screened recommendation is not refuted. KB/Conductor unavailable; primary/repository/Mathlib fallback used. The generic Smart proof and matched SafeCurves benchmark remain open. [pcc:unverified]